

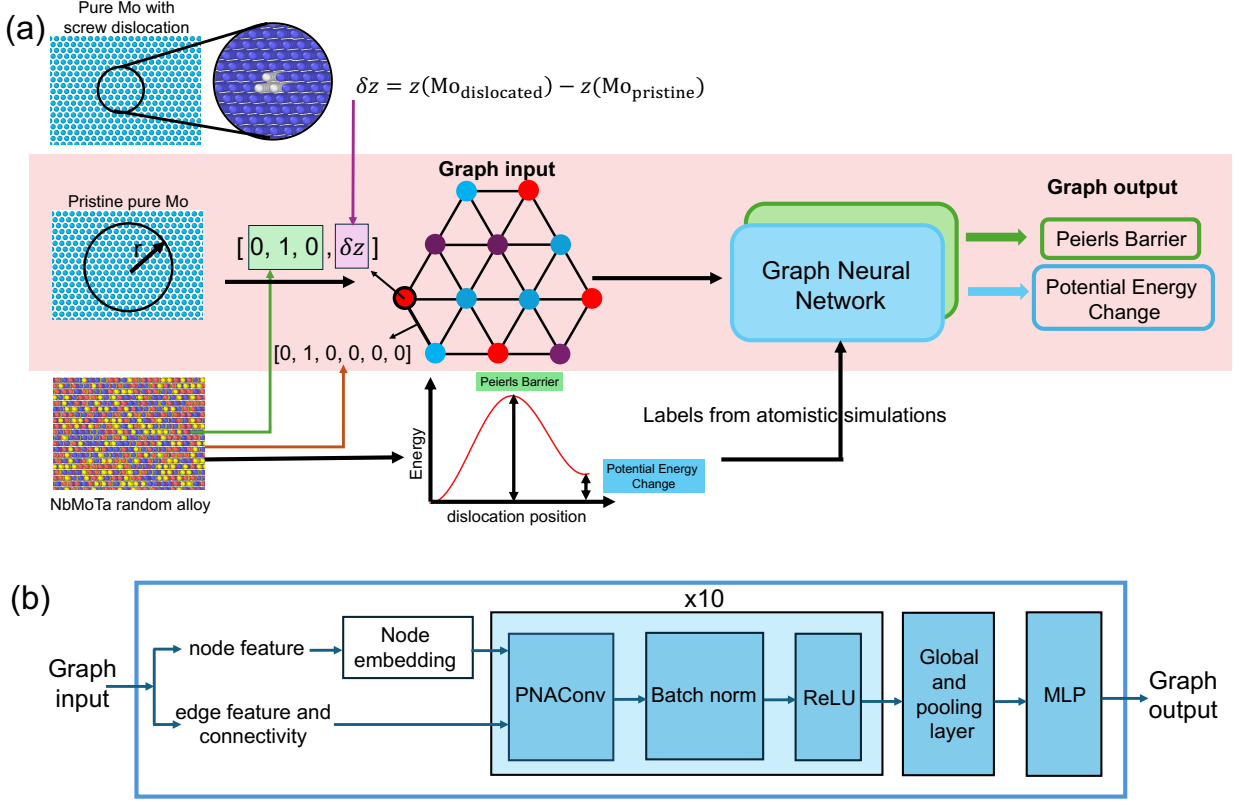


Figure 1: (a) **Overview of the workflow** used in this work to train GNN models for an end-to-end prediction of the Peierls barrier and potential energy changes. The process begins with the generation of initial and final dislocation structures in random BCC multi-component alloys, which are then minimized to compute the initial and final potential energies. Next, NEB simulations are performed to determine the minimum energy path and calculate the Peierls barrier. The input to the machine learning model is the graph representation of the alloys, with nodes encoding spatial and chemical information, and edges representing bond types. Two supervised GNN models are trained to predict graph-level labels: the Peierls barrier and potential energy change. (b) **GNN architecture.** The graph input is first passed into an input block to enlarge the dimension of node features. The node embedding, edge features and connectivity of input graphs are then input to the message passing block where the information from the neighbors of each node in the graph is aggregated to update the hidden features of the nodes. The output of the message passing block is then input into a global pooling later which outputs a graph-level embedding by adding node embeddings across the node, and it is connected to a multilayer perceptron (MLP) that returns a predicted graph label, i.e. Peierls barrier or potential energy change.

2.1.1 Peierls barrier

Random fluctuations in the solute environment surrounding the screw dislocation play a crucial role in determining the Peierls barrier, introducing a degree of complexity that must be captured for accurate predictions. In this section, we assess how well our GNN model predicts the Peierls barrier for various random compositions within the test set. To ensure the model’s generalizability, the compositions in the test set are strategically distributed to cover a broad range of the compositional space, including the equimolar composition $\text{Nb}_{33}\text{Mo}_{33}\text{Ta}_{33}$ and binary systems. This comprehensive distribution allows us to verify that the model can accurately capture both the complexity of ternary alloys and the simpler binary ones, which is critical for extending the model to new, unseen compositions. Moreover, each composition consists of 50 realizations of random solute configurations, ensuring a robust evaluation of the model’s predictive capabilities across different atomic arrangements.

We evaluate the model’s performance by comparing the predicted Peierls barrier values with the ground truth data, as illustrated in Figure 2(a). The model achieves a low mean absolute error of 37 meV, indicating that it has effectively captured the fundamental physical mechanisms driving dislocation motion. This accuracy,